

Array Manipulation

Starting with a 1-indexed array of zeros and a list of operations, for each operation add a value to each array element between two given indices, inclusive. Once all operations have been performed, return the maximum value in the array.

Example

```
n = 10
queries = [[1, 5, 3], [4, 8, 7], [6, 9, 1]]
```

Queries are interpreted as follows:

a	b	k
1	5	3
4	8	7
6	9	1

Add the values of *k* between the indices *a* and *b* inclusive:

0	0	0	0	0	0	0	0	0	0
3	3	3	3	3	0	0	0	0	0
3	3	3	10	10	7	7	7	0	0
3	3	3	10	10	8	8	8	1	0

The largest value is **10** after all operations are performed.

Function Description

Complete the function *arrayManipulation* with the following parameters:

- *int n*: the number of elements in the array
- *int queries*[*q*][3]: a two dimensional array of queries where each *queries*[*i*] contains three integers, *a*, *b*, and *k*.

Returns

- *int*: the maximum value in the resultant array

Input Format

The first line contains two space-separated integers *n* and *q*, the size of the array and the number of queries.
Each of the next *q* lines contains three space-separated integers *a*, *b* and *k*, the left index, right index and number to add.

Constraints

- $3 \leq n \leq 10^7$
- $1 \leq m \leq 2 * 10^5$

- $1 \leq a \leq b \leq n$

- $0 \leq k \leq 10^9$

Sample Input

STDIN	Function
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5 3	arr[] size n = 5, queries[] size q = 3
1 2 100	queries = [[1, 2, 100], [2, 5, 100], [3, 4, 100]]
2 5 100	
3 4 100	

Sample Output

200

Explanation

After the first update the list is 100 100 0 0 0.

After the second update list is 100 200 100 100 100.

After the third update list is 100 200 200 200 100.

The maximum value is 200.